A.V.I.A. (Autonomous Vital Intelligence Agent)

Project Plan - 2026 NASA Space Apps Challenge

1. Executive Summary

A.V.I.A. is an offline-first, AI-driven health monitoring system designed to mitigate the physiological and psychological risks of long-duration spaceflight. The system tracks cardiovascular adaptations, bone loss, immune dysregulation, and behavioral health, providing proactive countermeasures natively on local hardware to ensure strict data privacy and continuous operation without Earth-based cloud connectivity.

2. System Architecture

The solution operates across two primary interfaces, communicating via an encrypted local sync protocol (Bluetooth LE / Local Wi-Fi) to ensure data never leaks outside the closed environment.

A.V.I.A. Pulse (The Edge Interface)

- **Hardware Form Factor:** Smartwatch / Hand-held Smart Device
- **Primary Function:** Continuous vitals tracking (HR, SpO2, EDA), quick-action symptom logging, and critical alert generation.
- **Tech Stack:** TensorFlow Lite / ONNX for localized threshold-based anomaly detection.
- **UI/UX Design:** High-tactility design with large touch targets, swipe-heavy navigation, utilitarian typography (Helvetica/Inter), and strict functional color coding (Blue for nominal, Amber for warning, Red for critical).

A.V.I.A. Cockpit (The Control Center)

- **Hardware Form Factor:** Desktop Software
- **Primary Function:** Comprehensive holistic dashboard, deep-dive health analytics, mission control setup, and offline AI processing.
- **Tech Stack:** React (Frontend), Fastify + Node.js (API routing), Python (Heavy ML processing), PostgreSQL with Row Level Security (RLS).
- **3D Digital Twin Engine:** Built via React Three Fiber to render a low-poly interactive 3D human body.
    - **Multi-Layered Toggles:** Skeletal View (mechanical load mapping), Cardiovascular View (fluid shift visualization), and Dermal View (immune rash pinpointing).
    - **Contextual Deep-Drill Panels:** Clicking an affected body part opens telemetry data and actionable countermeasures.

    - **Predictive Timeline Slider:** Projects 30-60 days of health degradation using LSTM models based on current biometric trajectories.

3. Data Privacy & AI Strategy

- **Federated Learning:** Pre-trained baseline models refine themselves locally on the astronaut's device. No personal health data is uploaded to a central server.
- **Biometric & Psychological Agents:** LSTM models predict cardiovascular and bone density risks from time-series telemetry. NLP Transformer models run sentiment analysis on local voice/text journal entries to catch early markers of psychiatric strain.
- **External Datasets for Initial Training:** NASA OSDR, NASA LSDA, WESAD, PhysioNet, and MIMIC.

4. Execution Plan & Work Breakdown

The hackathon workflow will run in parallel to maximize efficiency before submission:

- **React Frontend & 3D Prototyping:** The Frontend Team will handle the A.V.I.A. Cockpit dashboard grids, the functional NASA-oriented UI styling, and the React Three Fiber digital twin integration.
- **Backend, API, & Local Sync Protocol:** The Backend Team will establish the Fastify/Node.js architecture, secure the local PostgreSQL RLS database, and map the offline telemetry pipeline.
- **AI/ML Edge Deployment:** The AI/ML Team will focus on training the LSTM/NLP models using the external datasets and quantizing them for TensorFlow Lite deployment on the Pulse edge device.